" বিস্মিল্লাহির রাহ্মানির রাহীম "

GAMMA & BETA FUNCTIONS:

The gamma func first came into view in connection with the interpolation problems for factorials. It was solved by Euler. The modern notation for gamma and beta func is due to legendre.

GAMMA FUNCTION:

The definite integral $\int_0^\infty e^{-x} x^{n-1} dx$ is called second Eulerian integral or G.F. and is denoted by $\Gamma(n)$, that is

$$\Gamma n = \int_0^\infty e^{-x} x^{n-1} dx$$

Here n is +ve but need not be integer.

BETA FUNCTION:

The definite integral $\int_0^1 x^{m-1} (1-x)^{n-1} dx$ is called the first Eulerian integral or B.F. and is denoted by $B(m,n)$. Here

$$B(m,n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

Here m and n are positive but need not be integer.

Ex: $\int_0^\infty \frac{dy}{1+y^4}$

Sol^n: $\int_0^\infty \frac{dy}{1+y^4}$

let, $y^4 = x$

$\Rightarrow y = x^{1/4}$

$\Rightarrow dy = \frac{1}{4} x^{1/4-1} dx$

$= \frac{1}{4} x^{-3/4} dx$

when $y = 0,\ x = 0$

$y = \infty,\ x = \infty$

$$= \frac{1}{4}\int_0^\infty \frac{x^{-3/4}}{1+x}\,dx \quad (1)$$

$$= \frac{1}{4}\int_0^\infty \frac{x^{\frac{1}{4}-1}}{1+x}\,dx$$

$$= \frac{1}{4}\,\frac{\pi}{\sin\left(\frac{1}{4}\cdot\pi\right)}$$

$$= \frac{1}{4}\,\frac{\pi}{\sin\left(\frac{\pi}{4}\right)}$$

$$= \frac{1}{4}\,\frac{\pi}{1/\sqrt{2}}$$

$$= \frac{\sqrt{2}\,\pi}{4}$$

Rule: $\int_0^\infty \frac{x^{m-1}}{1+x}\,dx = \frac{\pi}{\sin m\pi}$

Ex: Find ⓐ $\Gamma(-\frac{1}{2})$ & ⓑ $\Gamma(-5/2)$.

Solⁿ: we use generalization to negative values defined by

$$\Gamma(n) = \frac{\Gamma(n+1)}{n} \quad \text{——— ①}$$

ⓐ putting $n = -1/2$ in ① we get,

$$\Gamma(-1/2) = \frac{\Gamma(1/2)}{-1/2}$$

$$= \frac{\sqrt{\pi}}{-1/2}$$

$$= -2\sqrt{\pi}.$$

ⓑ putting $n = -5/2$ in ① we get,

$$\Gamma(-5/2) = \frac{\Gamma(-5/2+1)}{-5/2}$$

$$= \frac{\Gamma(-3/2)}{-5/2} \quad \text{——— (II)}$$

Now, $\Gamma(-3/2) = \frac{\Gamma(-3/2+1)}{-3/2} = \frac{\Gamma(-1/2)}{-3/2} = \frac{-2\sqrt{\pi}}{-3/2} = \frac{4\sqrt{\pi}}{3}$

From (II) we have,

$$\Gamma(-5/2) = \frac{4\sqrt{\pi}}{3} \times \frac{-2}{5} = -\frac{8}{15}\sqrt{\pi}.$$

⊞ Show that, $\int_0^\infty e^{-x^2} x^{\alpha^2} dx = \frac{1}{2} \cdot \Gamma\left(\frac{\alpha^2+1}{2}\right)$

Sol^n: let $I = \int_0^\infty e^{-x^2} x^{\alpha^2} dx$

put, $x^2 = y$

$\Rightarrow 2x\,dx = dy$

$\Rightarrow dx = \frac{1}{2x} dy$

$= \frac{dy}{2y^{1/2}}$

$\therefore dx = \frac{y^{-1/2}}{2} dy$

when $x=0$ then $y=0$

" $x=\infty$ then $y=\infty$

$= \int_0^\infty e^{-y} y^{\frac{1}{2}\alpha^2} \frac{1}{2} y^{-\frac{1}{2}} dy$

$= \frac{1}{2}\int_0^\infty e^{-y} y^{\frac{1}{2}\alpha^2 - \frac{1}{2}} dy$

$= \frac{1}{2}\int_0^\infty e^{-y} y^{\frac{1}{2}\alpha^2 + \frac{1}{2} - 1} dy$

$= \frac{1}{2}\int_0^\infty e^{-y} y^{\left(\frac{\alpha^2+1}{2} - 1\right)} dy$

$= \frac{1}{2} \Gamma\left(\frac{\alpha^2+1}{2}\right)$

$\therefore \int_0^\infty e^{-x^2} x^{\alpha^2} dx = \frac{1}{2}\Gamma\left(\frac{\alpha^2+1}{2}\right)$

Rule: $\int_0^\infty e^{-y} y^{m-1} dy = \Gamma m$